NCERT CLASS 10
60 Questions Per Topic

Based on the refined 2026–27 curriculum map created for this project.

Each curriculum topic below contains exactly 60 questions: understanding, examples/classification, comparison, application, reasoning/error analysis, and higher-order activity.

These are question-bank items derived from the topic descriptions in the supplied curriculum map. Validate against the exact school/textbook edition before high-stakes use.

MATHEMATICS — 1. Real Numbers

Coverage: ■ Euclid division lemma; fundamental theorem of arithmetic; irrational numbers; applications.

Understanding

1. What is ■ Euclid division lemma, and what are its main features? Use a simple example.

2. What is fundamental theorem of arithmetic, and what are its main features? Use a school-life example.

3. What is irrational numbers, and what are its main features? Use a real-world example.

4. What is applications., and what are its main features? Show the idea with a diagram where appropriate.

5. What is ■ Euclid division lemma, and what are its main features? Explain in your own words.

6. What is fundamental theorem of arithmetic, and what are its main features? Give a step-by-step response.

7. What is irrational numbers, and what are its main features? Mention one common misconception.

8. What is applications., and what are its main features? State one practical application.

9. What is ■ Euclid division lemma, and what are its main features? Justify your response.

10. What is fundamental theorem of arithmetic, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to irrational numbers and explain each. Use a simple example.

12. Give two examples related to applications. and explain each. Use a school-life example.

13. Give two examples related to ■ Euclid division lemma and explain each. Use a real-world example.

14. Give two examples related to fundamental theorem of arithmetic and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to irrational numbers and explain each. Explain in your own words.

16. Give two examples related to applications. and explain each. Give a step-by-step response.

17. Give two examples related to ■ Euclid division lemma and explain each. Mention one common misconception.

18. Give two examples related to fundamental theorem of arithmetic and explain each. State one practical application.

19. Give two examples related to irrational numbers and explain each. Justify your response.

20. Give two examples related to applications. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Euclid division lemma related to fundamental theorem of arithmetic? Compare the two ideas. Use a simple example.

22. How is fundamental theorem of arithmetic related to irrational numbers? Compare the two ideas. Use a school-life example.

23. How is irrational numbers related to applications.? Compare the two ideas. Use a real-world example.

24. How is applications. related to ■ Euclid division lemma? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Euclid division lemma related to fundamental theorem of arithmetic? Compare the two ideas. Explain in your own words.

26. How is fundamental theorem of arithmetic related to irrational numbers? Compare the two ideas. Give a step-by-step response.

27. How is irrational numbers related to applications.? Compare the two ideas. Mention one common misconception.

28. How is applications. related to ■ Euclid division lemma? Compare the two ideas. State one practical application.

29. How is ■ Euclid division lemma related to fundamental theorem of arithmetic? Compare the two ideas. Justify your response.

30. How is fundamental theorem of arithmetic related to irrational numbers? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply irrational numbers to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply applications. to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply ■ Euclid division lemma to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply fundamental theorem of arithmetic to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply irrational numbers to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply applications. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Euclid division lemma to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply fundamental theorem of arithmetic to a realistic situation. What would you do or calculate? State one practical application.

39. Apply irrational numbers to a realistic situation. What would you do or calculate? Justify your response.

40. Apply applications. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Euclid division lemma. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of fundamental theorem of arithmetic. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of irrational numbers. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of applications.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of ■ Euclid division lemma. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of fundamental theorem of arithmetic. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of irrational numbers. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of applications.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Euclid division lemma. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of fundamental theorem of arithmetic. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show irrational numbers. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Euclid division lemma. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show fundamental theorem of arithmetic. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show irrational numbers. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Euclid division lemma. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show fundamental theorem of arithmetic. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show irrational numbers. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Include the relevant rule or principle.

MATHEMATICS — 2. Polynomials

Coverage: ■ Zeroes; coefficients; relationship; division algorithm; algebraic identities.

Understanding

1. What is ■ Zeroes, and what are its main features? Use a simple example.

2. What is coefficients, and what are its main features? Use a school-life example.

3. What is relationship, and what are its main features? Use a real-world example.

4. What is division algorithm, and what are its main features? Show the idea with a diagram where appropriate.

5. What is algebraic identities., and what are its main features? Explain in your own words.

6. What is ■ Zeroes, and what are its main features? Give a step-by-step response.

7. What is coefficients, and what are its main features? Mention one common misconception.

8. What is relationship, and what are its main features? State one practical application.

9. What is division algorithm, and what are its main features? Justify your response.

10. What is algebraic identities., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Zeroes and explain each. Use a simple example.

12. Give two examples related to coefficients and explain each. Use a school-life example.

13. Give two examples related to relationship and explain each. Use a real-world example.

14. Give two examples related to division algorithm and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to algebraic identities. and explain each. Explain in your own words.

16. Give two examples related to ■ Zeroes and explain each. Give a step-by-step response.

17. Give two examples related to coefficients and explain each. Mention one common misconception.

18. Give two examples related to relationship and explain each. State one practical application.

19. Give two examples related to division algorithm and explain each. Justify your response.

20. Give two examples related to algebraic identities. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Zeroes related to coefficients? Compare the two ideas. Use a simple example.

22. How is coefficients related to relationship? Compare the two ideas. Use a school-life example.

23. How is relationship related to division algorithm? Compare the two ideas. Use a real-world example.

24. How is division algorithm related to algebraic identities.? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is algebraic identities. related to ■ Zeroes? Compare the two ideas. Explain in your own words.

26. How is ■ Zeroes related to coefficients? Compare the two ideas. Give a step-by-step response.

27. How is coefficients related to relationship? Compare the two ideas. Mention one common misconception.

28. How is relationship related to division algorithm? Compare the two ideas. State one practical application.

29. How is division algorithm related to algebraic identities.? Compare the two ideas. Justify your response.

30. How is algebraic identities. related to ■ Zeroes? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Zeroes to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply coefficients to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply relationship to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply division algorithm to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply algebraic identities. to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply ■ Zeroes to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply coefficients to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply relationship to a realistic situation. What would you do or calculate? State one practical application.

39. Apply division algorithm to a realistic situation. What would you do or calculate? Justify your response.

40. Apply algebraic identities. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Zeroes. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of coefficients. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of relationship. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of division algorithm. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of algebraic identities.. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of ■ Zeroes. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of coefficients. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of relationship. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of division algorithm. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of algebraic identities.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Zeroes. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show coefficients. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show relationship. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show division algorithm. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show algebraic identities.. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Zeroes. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show coefficients. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show relationship. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show division algorithm. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show algebraic identities.. Include the relevant rule or principle.

MATHEMATICS — 3. Pair of Linear Equations in Two Variables

Coverage: ■ Graphical and algebraic methods; substitution; elimination; cross-multiplication.

Understanding

1. What is ■ Graphical and algebraic methods, and what are its main features? Use a simple example.

2. What is substitution, and what are its main features? Use a school-life example.

3. What is elimination, and what are its main features? Use a real-world example.

4. What is cross-multiplication., and what are its main features? Show the idea with a diagram where appropriate.

5. What is ■ Graphical and algebraic methods, and what are its main features? Explain in your own words.

6. What is substitution, and what are its main features? Give a step-by-step response.

7. What is elimination, and what are its main features? Mention one common misconception.

8. What is cross-multiplication., and what are its main features? State one practical application.

9. What is ■ Graphical and algebraic methods, and what are its main features? Justify your response.

10. What is substitution, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to elimination and explain each. Use a simple example.

12. Give two examples related to cross-multiplication. and explain each. Use a school-life example.

13. Give two examples related to ■ Graphical and algebraic methods and explain each. Use a real-world example.

14. Give two examples related to substitution and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to elimination and explain each. Explain in your own words.

16. Give two examples related to cross-multiplication. and explain each. Give a step-by-step response.

17. Give two examples related to ■ Graphical and algebraic methods and explain each. Mention one common misconception.

18. Give two examples related to substitution and explain each. State one practical application.

19. Give two examples related to elimination and explain each. Justify your response.

20. Give two examples related to cross-multiplication. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Graphical and algebraic methods related to substitution? Compare the two ideas. Use a simple example.

22. How is substitution related to elimination? Compare the two ideas. Use a school-life example.

23. How is elimination related to cross-multiplication.? Compare the two ideas. Use a real-world example.

24. How is cross-multiplication. related to ■ Graphical and algebraic methods? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Graphical and algebraic methods related to substitution? Compare the two ideas. Explain in your own words.

26. How is substitution related to elimination? Compare the two ideas. Give a step-by-step response.

27. How is elimination related to cross-multiplication.? Compare the two ideas. Mention one common misconception.

28. How is cross-multiplication. related to ■ Graphical and algebraic methods? Compare the two ideas. State one practical application.

29. How is ■ Graphical and algebraic methods related to substitution? Compare the two ideas. Justify your response.

30. How is substitution related to elimination? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply elimination to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply cross-multiplication. to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply ■ Graphical and algebraic methods to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply substitution to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply elimination to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply cross-multiplication. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Graphical and algebraic methods to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply substitution to a realistic situation. What would you do or calculate? State one practical application.

39. Apply elimination to a realistic situation. What would you do or calculate? Justify your response.

40. Apply cross-multiplication. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Graphical and algebraic methods. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of substitution. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of elimination. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of cross-multiplication.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of ■ Graphical and algebraic methods. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of substitution. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of elimination. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of cross-multiplication.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Graphical and algebraic methods. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of substitution. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show elimination. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show cross-multiplication.. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Graphical and algebraic methods. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show substitution. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show elimination. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show cross-multiplication.. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Graphical and algebraic methods. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show substitution. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show elimination. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show cross-multiplication.. Include the relevant rule or principle.

MATHEMATICS — 4. Quadratic Equations

Coverage: ■ Factorisation; completing square; quadratic formula; nature of roots.

Understanding

1. What is ■ Factorisation, and what are its main features? Use a simple example.

2. What is completing square, and what are its main features? Use a school-life example.

3. What is quadratic formula, and what are its main features? Use a real-world example.

4. What is nature of roots., and what are its main features? Show the idea with a diagram where appropriate.

5. What is ■ Factorisation, and what are its main features? Explain in your own words.

6. What is completing square, and what are its main features? Give a step-by-step response.

7. What is quadratic formula, and what are its main features? Mention one common misconception.

8. What is nature of roots., and what are its main features? State one practical application.

9. What is ■ Factorisation, and what are its main features? Justify your response.

10. What is completing square, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to quadratic formula and explain each. Use a simple example.

12. Give two examples related to nature of roots. and explain each. Use a school-life example.

13. Give two examples related to ■ Factorisation and explain each. Use a real-world example.

14. Give two examples related to completing square and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to quadratic formula and explain each. Explain in your own words.

16. Give two examples related to nature of roots. and explain each. Give a step-by-step response.

17. Give two examples related to ■ Factorisation and explain each. Mention one common misconception.

18. Give two examples related to completing square and explain each. State one practical application.

19. Give two examples related to quadratic formula and explain each. Justify your response.

20. Give two examples related to nature of roots. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Factorisation related to completing square? Compare the two ideas. Use a simple example.

22. How is completing square related to quadratic formula? Compare the two ideas. Use a school-life example.

23. How is quadratic formula related to nature of roots.? Compare the two ideas. Use a real-world example.

24. How is nature of roots. related to ■ Factorisation? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Factorisation related to completing square? Compare the two ideas. Explain in your own words.

26. How is completing square related to quadratic formula? Compare the two ideas. Give a step-by-step response.

27. How is quadratic formula related to nature of roots.? Compare the two ideas. Mention one common misconception.

28. How is nature of roots. related to ■ Factorisation? Compare the two ideas. State one practical application.

29. How is ■ Factorisation related to completing square? Compare the two ideas. Justify your response.

30. How is completing square related to quadratic formula? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply quadratic formula to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply nature of roots. to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply ■ Factorisation to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply completing square to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply quadratic formula to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply nature of roots. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Factorisation to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply completing square to a realistic situation. What would you do or calculate? State one practical application.

39. Apply quadratic formula to a realistic situation. What would you do or calculate? Justify your response.

40. Apply nature of roots. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Factorisation. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of completing square. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of quadratic formula. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of nature of roots.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of ■ Factorisation. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of completing square. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of quadratic formula. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of nature of roots.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Factorisation. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of completing square. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show quadratic formula. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show nature of roots.. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Factorisation. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show completing square. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show quadratic formula. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show nature of roots.. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Factorisation. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show completing square. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show quadratic formula. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show nature of roots.. Include the relevant rule or principle.

MATHEMATICS — 5. Arithmetic Progressions

Coverage: ■ Sequences; nth term; sum of n terms; applications.

Understanding

1. What is ■ Sequences, and what are its main features? Use a simple example.

2. What is nth term, and what are its main features? Use a school-life example.

3. What is sum of n terms, and what are its main features? Use a real-world example.

4. What is applications., and what are its main features? Show the idea with a diagram where appropriate.

5. What is ■ Sequences, and what are its main features? Explain in your own words.

6. What is nth term, and what are its main features? Give a step-by-step response.

7. What is sum of n terms, and what are its main features? Mention one common misconception.

8. What is applications., and what are its main features? State one practical application.

9. What is ■ Sequences, and what are its main features? Justify your response.

10. What is nth term, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to sum of n terms and explain each. Use a simple example.

12. Give two examples related to applications. and explain each. Use a school-life example.

13. Give two examples related to ■ Sequences and explain each. Use a real-world example.

14. Give two examples related to nth term and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to sum of n terms and explain each. Explain in your own words.

16. Give two examples related to applications. and explain each. Give a step-by-step response.

17. Give two examples related to ■ Sequences and explain each. Mention one common misconception.

18. Give two examples related to nth term and explain each. State one practical application.

19. Give two examples related to sum of n terms and explain each. Justify your response.

20. Give two examples related to applications. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Sequences related to nth term? Compare the two ideas. Use a simple example.

22. How is nth term related to sum of n terms? Compare the two ideas. Use a school-life example.

23. How is sum of n terms related to applications.? Compare the two ideas. Use a real-world example.

24. How is applications. related to ■ Sequences? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Sequences related to nth term? Compare the two ideas. Explain in your own words.

26. How is nth term related to sum of n terms? Compare the two ideas. Give a step-by-step response.

27. How is sum of n terms related to applications.? Compare the two ideas. Mention one common misconception.

28. How is applications. related to ■ Sequences? Compare the two ideas. State one practical application.

29. How is ■ Sequences related to nth term? Compare the two ideas. Justify your response.

30. How is nth term related to sum of n terms? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply sum of n terms to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply applications. to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply ■ Sequences to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply nth term to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply sum of n terms to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply applications. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Sequences to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply nth term to a realistic situation. What would you do or calculate? State one practical application.

39. Apply sum of n terms to a realistic situation. What would you do or calculate? Justify your response.

40. Apply applications. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Sequences. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of nth term. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of sum of n terms. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of applications.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of ■ Sequences. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of nth term. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of sum of n terms. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of applications.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Sequences. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of nth term. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show sum of n terms. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Sequences. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show nth term. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show sum of n terms. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Sequences. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show nth term. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show sum of n terms. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Include the relevant rule or principle.

MATHEMATICS — 6. Triangles

Coverage: ■ Similarity; Basic Proportionality Theorem; Pythagoras theorem; applications.

Understanding

1. What is ■ Similarity, and what are its main features? Use a simple example.

2. What is Basic Proportionality Theorem, and what are its main features? Use a school-life example.

3. What is Pythagoras theorem, and what are its main features? Use a real-world example.

4. What is applications., and what are its main features? Show the idea with a diagram where appropriate.

5. What is ■ Similarity, and what are its main features? Explain in your own words.

6. What is Basic Proportionality Theorem, and what are its main features? Give a step-by-step response.

7. What is Pythagoras theorem, and what are its main features? Mention one common misconception.

8. What is applications., and what are its main features? State one practical application.

9. What is ■ Similarity, and what are its main features? Justify your response.

10. What is Basic Proportionality Theorem, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to Pythagoras theorem and explain each. Use a simple example.

12. Give two examples related to applications. and explain each. Use a school-life example.

13. Give two examples related to ■ Similarity and explain each. Use a real-world example.

14. Give two examples related to Basic Proportionality Theorem and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to Pythagoras theorem and explain each. Explain in your own words.

16. Give two examples related to applications. and explain each. Give a step-by-step response.

17. Give two examples related to ■ Similarity and explain each. Mention one common misconception.

18. Give two examples related to Basic Proportionality Theorem and explain each. State one practical application.

19. Give two examples related to Pythagoras theorem and explain each. Justify your response.

20. Give two examples related to applications. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Similarity related to Basic Proportionality Theorem? Compare the two ideas. Use a simple example.

22. How is Basic Proportionality Theorem related to Pythagoras theorem? Compare the two ideas. Use a school-life example.

23. How is Pythagoras theorem related to applications.? Compare the two ideas. Use a real-world example.

24. How is applications. related to ■ Similarity? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Similarity related to Basic Proportionality Theorem? Compare the two ideas. Explain in your own words.

26. How is Basic Proportionality Theorem related to Pythagoras theorem? Compare the two ideas. Give a step-by-step response.

27. How is Pythagoras theorem related to applications.? Compare the two ideas. Mention one common misconception.

28. How is applications. related to ■ Similarity? Compare the two ideas. State one practical application.

29. How is ■ Similarity related to Basic Proportionality Theorem? Compare the two ideas. Justify your response.

30. How is Basic Proportionality Theorem related to Pythagoras theorem? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply Pythagoras theorem to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply applications. to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply ■ Similarity to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply Basic Proportionality Theorem to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply Pythagoras theorem to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply applications. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Similarity to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply Basic Proportionality Theorem to a realistic situation. What would you do or calculate? State one practical application.

39. Apply Pythagoras theorem to a realistic situation. What would you do or calculate? Justify your response.

40. Apply applications. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Similarity. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of Basic Proportionality Theorem. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of Pythagoras theorem. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of applications.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of ■ Similarity. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of Basic Proportionality Theorem. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of Pythagoras theorem. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of applications.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Similarity. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of Basic Proportionality Theorem. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show Pythagoras theorem. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Similarity. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show Basic Proportionality Theorem. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show Pythagoras theorem. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Similarity. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show Basic Proportionality Theorem. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show Pythagoras theorem. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Include the relevant rule or principle.

MATHEMATICS — 7. Coordinate Geometry

Coverage: ■ Distance; section formula; area of triangle.

Understanding

1. What is ■ Distance, and what are its main features? Use a simple example.

2. What is section formula, and what are its main features? Use a school-life example.

3. What is area of triangle., and what are its main features? Use a real-world example.

4. What is ■ Distance, and what are its main features? Show the idea with a diagram where appropriate.

5. What is section formula, and what are its main features? Explain in your own words.

6. What is area of triangle., and what are its main features? Give a step-by-step response.

7. What is ■ Distance, and what are its main features? Mention one common misconception.

8. What is section formula, and what are its main features? State one practical application.

9. What is area of triangle., and what are its main features? Justify your response.

10. What is ■ Distance, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to section formula and explain each. Use a simple example.

12. Give two examples related to area of triangle. and explain each. Use a school-life example.

13. Give two examples related to ■ Distance and explain each. Use a real-world example.

14. Give two examples related to section formula and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to area of triangle. and explain each. Explain in your own words.

16. Give two examples related to ■ Distance and explain each. Give a step-by-step response.

17. Give two examples related to section formula and explain each. Mention one common misconception.

18. Give two examples related to area of triangle. and explain each. State one practical application.

19. Give two examples related to ■ Distance and explain each. Justify your response.

20. Give two examples related to section formula and explain each. Include the relevant rule or principle.

Comparison

21. How is area of triangle. related to ■ Distance? Compare the two ideas. Use a simple example.

22. How is ■ Distance related to section formula? Compare the two ideas. Use a school-life example.

23. How is section formula related to area of triangle.? Compare the two ideas. Use a real-world example.

24. How is area of triangle. related to ■ Distance? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Distance related to section formula? Compare the two ideas. Explain in your own words.

26. How is section formula related to area of triangle.? Compare the two ideas. Give a step-by-step response.

27. How is area of triangle. related to ■ Distance? Compare the two ideas. Mention one common misconception.

28. How is ■ Distance related to section formula? Compare the two ideas. State one practical application.

29. How is section formula related to area of triangle.? Compare the two ideas. Justify your response.

30. How is area of triangle. related to ■ Distance? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Distance to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply section formula to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply area of triangle. to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply ■ Distance to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply section formula to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply area of triangle. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Distance to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply section formula to a realistic situation. What would you do or calculate? State one practical application.

39. Apply area of triangle. to a realistic situation. What would you do or calculate? Justify your response.

40. Apply ■ Distance to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of section formula. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of area of triangle.. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of ■ Distance. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of section formula. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of area of triangle.. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of ■ Distance. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of section formula. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of area of triangle.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Distance. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of section formula. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show area of triangle.. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Distance. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show section formula. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show area of triangle.. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Distance. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show section formula. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show area of triangle.. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Distance. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show section formula. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show area of triangle.. Include the relevant rule or principle.

MATHEMATICS — 8. Introduction to Trigonometry

Coverage: ■ Trigonometric ratios; identities; values and relationships.

Understanding

1. What is ■ Trigonometric ratios, and what are its main features? Use a simple example.

2. What is identities, and what are its main features? Use a school-life example.

3. What is values and relationships., and what are its main features? Use a real-world example.

4. What is ■ Trigonometric ratios, and what are its main features? Show the idea with a diagram where appropriate.

5. What is identities, and what are its main features? Explain in your own words.

6. What is values and relationships., and what are its main features? Give a step-by-step response.

7. What is ■ Trigonometric ratios, and what are its main features? Mention one common misconception.

8. What is identities, and what are its main features? State one practical application.

9. What is values and relationships., and what are its main features? Justify your response.

10. What is ■ Trigonometric ratios, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to identities and explain each. Use a simple example.

12. Give two examples related to values and relationships. and explain each. Use a school-life example.

13. Give two examples related to ■ Trigonometric ratios and explain each. Use a real-world example.

14. Give two examples related to identities and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to values and relationships. and explain each. Explain in your own words.

16. Give two examples related to ■ Trigonometric ratios and explain each. Give a step-by-step response.

17. Give two examples related to identities and explain each. Mention one common misconception.

18. Give two examples related to values and relationships. and explain each. State one practical application.

19. Give two examples related to ■ Trigonometric ratios and explain each. Justify your response.

20. Give two examples related to identities and explain each. Include the relevant rule or principle.

Comparison

21. How is values and relationships. related to ■ Trigonometric ratios? Compare the two ideas. Use a simple example.

22. How is ■ Trigonometric ratios related to identities? Compare the two ideas. Use a school-life example.

23. How is identities related to values and relationships.? Compare the two ideas. Use a real-world example.

24. How is values and relationships. related to ■ Trigonometric ratios? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Trigonometric ratios related to identities? Compare the two ideas. Explain in your own words.

26. How is identities related to values and relationships.? Compare the two ideas. Give a step-by-step response.

27. How is values and relationships. related to ■ Trigonometric ratios? Compare the two ideas. Mention one common misconception.

28. How is ■ Trigonometric ratios related to identities? Compare the two ideas. State one practical application.

29. How is identities related to values and relationships.? Compare the two ideas. Justify your response.

30. How is values and relationships. related to ■ Trigonometric ratios? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Trigonometric ratios to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply identities to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply values and relationships. to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply ■ Trigonometric ratios to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply identities to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply values and relationships. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Trigonometric ratios to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply identities to a realistic situation. What would you do or calculate? State one practical application.

39. Apply values and relationships. to a realistic situation. What would you do or calculate? Justify your response.

40. Apply ■ Trigonometric ratios to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of identities. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of values and relationships.. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of ■ Trigonometric ratios. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of identities. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of values and relationships.. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of ■ Trigonometric ratios. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of identities. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of values and relationships.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Trigonometric ratios. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of identities. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show values and relationships.. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Trigonometric ratios. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show identities. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show values and relationships.. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Trigonometric ratios. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show identities. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show values and relationships.. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Trigonometric ratios. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show identities. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show values and relationships.. Include the relevant rule or principle.

MATHEMATICS — 9. Some Applications of Trigonometry

Coverage: ■ Heights and distances; angles of elevation/depression.

Understanding

1. What is ■ Heights and distances, and what are its main features? Use a simple example.

2. What is angles of elevation/depression., and what are its main features? Use a school-life example.

3. What is ■ Heights and distances, and what are its main features? Use a real-world example.

4. What is angles of elevation/depression., and what are its main features? Show the idea with a diagram where appropriate.

5. What is ■ Heights and distances, and what are its main features? Explain in your own words.

6. What is angles of elevation/depression., and what are its main features? Give a step-by-step response.

7. What is ■ Heights and distances, and what are its main features? Mention one common misconception.

8. What is angles of elevation/depression., and what are its main features? State one practical application.

9. What is ■ Heights and distances, and what are its main features? Justify your response.

10. What is angles of elevation/depression., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Heights and distances and explain each. Use a simple example.

12. Give two examples related to angles of elevation/depression. and explain each. Use a school-life example.

13. Give two examples related to ■ Heights and distances and explain each. Use a real-world example.

14. Give two examples related to angles of elevation/depression. and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to ■ Heights and distances and explain each. Explain in your own words.

16. Give two examples related to angles of elevation/depression. and explain each. Give a step-by-step response.

17. Give two examples related to ■ Heights and distances and explain each. Mention one common misconception.

18. Give two examples related to angles of elevation/depression. and explain each. State one practical application.

19. Give two examples related to ■ Heights and distances and explain each. Justify your response.

20. Give two examples related to angles of elevation/depression. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Heights and distances related to angles of elevation/depression.? Compare the two ideas. Use a simple example.

22. How is angles of elevation/depression. related to ■ Heights and distances? Compare the two ideas. Use a school-life example.

23. How is ■ Heights and distances related to angles of elevation/depression.? Compare the two ideas. Use a real-world example.

24. How is angles of elevation/depression. related to ■ Heights and distances? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Heights and distances related to angles of elevation/depression.? Compare the two ideas. Explain in your own words.

26. How is angles of elevation/depression. related to ■ Heights and distances? Compare the two ideas. Give a step-by-step response.

27. How is ■ Heights and distances related to angles of elevation/depression.? Compare the two ideas. Mention one common misconception.

28. How is angles of elevation/depression. related to ■ Heights and distances? Compare the two ideas. State one practical application.

29. How is ■ Heights and distances related to angles of elevation/depression.? Compare the two ideas. Justify your response.

30. How is angles of elevation/depression. related to ■ Heights and distances? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Heights and distances to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply angles of elevation/depression. to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply ■ Heights and distances to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply angles of elevation/depression. to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply ■ Heights and distances to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply angles of elevation/depression. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Heights and distances to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply angles of elevation/depression. to a realistic situation. What would you do or calculate? State one practical application.

39. Apply ■ Heights and distances to a realistic situation. What would you do or calculate? Justify your response.

40. Apply angles of elevation/depression. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Heights and distances. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of angles of elevation/depression.. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of ■ Heights and distances. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of angles of elevation/depression.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of ■ Heights and distances. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of angles of elevation/depression.. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of ■ Heights and distances. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of angles of elevation/depression.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Heights and distances. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of angles of elevation/depression.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Heights and distances. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show angles of elevation/depression.. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Heights and distances. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show angles of elevation/depression.. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Heights and distances. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show angles of elevation/depression.. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Heights and distances. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show angles of elevation/depression.. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Heights and distances. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show angles of elevation/depression.. Include the relevant rule or principle.

MATHEMATICS — 10. Circles

Coverage: ■ Tangents; tangent properties; theorem-based reasoning.

Understanding

1. What is ■ Tangents, and what are its main features? Use a simple example.

2. What is tangent properties, and what are its main features? Use a school-life example.

3. What is theorem-based reasoning., and what are its main features? Use a real-world example.

4. What is ■ Tangents, and what are its main features? Show the idea with a diagram where appropriate.

5. What is tangent properties, and what are its main features? Explain in your own words.

6. What is theorem-based reasoning., and what are its main features? Give a step-by-step response.

7. What is ■ Tangents, and what are its main features? Mention one common misconception.

8. What is tangent properties, and what are its main features? State one practical application.

9. What is theorem-based reasoning., and what are its main features? Justify your response.

10. What is ■ Tangents, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to tangent properties and explain each. Use a simple example.

12. Give two examples related to theorem-based reasoning. and explain each. Use a school-life example.

13. Give two examples related to ■ Tangents and explain each. Use a real-world example.

14. Give two examples related to tangent properties and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to theorem-based reasoning. and explain each. Explain in your own words.

16. Give two examples related to ■ Tangents and explain each. Give a step-by-step response.

17. Give two examples related to tangent properties and explain each. Mention one common misconception.

18. Give two examples related to theorem-based reasoning. and explain each. State one practical application.

19. Give two examples related to ■ Tangents and explain each. Justify your response.

20. Give two examples related to tangent properties and explain each. Include the relevant rule or principle.

Comparison

21. How is theorem-based reasoning. related to ■ Tangents? Compare the two ideas. Use a simple example.

22. How is ■ Tangents related to tangent properties? Compare the two ideas. Use a school-life example.

23. How is tangent properties related to theorem-based reasoning.? Compare the two ideas. Use a real-world example.

24. How is theorem-based reasoning. related to ■ Tangents? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Tangents related to tangent properties? Compare the two ideas. Explain in your own words.

26. How is tangent properties related to theorem-based reasoning.? Compare the two ideas. Give a step-by-step response.

27. How is theorem-based reasoning. related to ■ Tangents? Compare the two ideas. Mention one common misconception.

28. How is ■ Tangents related to tangent properties? Compare the two ideas. State one practical application.

29. How is tangent properties related to theorem-based reasoning.? Compare the two ideas. Justify your response.

30. How is theorem-based reasoning. related to ■ Tangents? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Tangents to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply tangent properties to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply theorem-based reasoning. to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply ■ Tangents to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply tangent properties to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply theorem-based reasoning. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Tangents to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply tangent properties to a realistic situation. What would you do or calculate? State one practical application.

39. Apply theorem-based reasoning. to a realistic situation. What would you do or calculate? Justify your response.

40. Apply ■ Tangents to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of tangent properties. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of theorem-based reasoning.. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of ■ Tangents. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of tangent properties. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of theorem-based reasoning.. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of ■ Tangents. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of tangent properties. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of theorem-based reasoning.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Tangents. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of tangent properties. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show theorem-based reasoning.. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Tangents. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show tangent properties. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show theorem-based reasoning.. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Tangents. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show tangent properties. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show theorem-based reasoning.. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Tangents. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show tangent properties. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show theorem-based reasoning.. Include the relevant rule or principle.

MATHEMATICS — 11. Areas Related to Circles

Coverage: ■ Sectors; segments; combinations of plane figures.

Understanding

1. What is ■ Sectors, and what are its main features? Use a simple example.

2. What is segments, and what are its main features? Use a school-life example.

3. What is combinations of plane figures., and what are its main features? Use a real-world example.

4. What is ■ Sectors, and what are its main features? Show the idea with a diagram where appropriate.

5. What is segments, and what are its main features? Explain in your own words.

6. What is combinations of plane figures., and what are its main features? Give a step-by-step response.

7. What is ■ Sectors, and what are its main features? Mention one common misconception.

8. What is segments, and what are its main features? State one practical application.

9. What is combinations of plane figures., and what are its main features? Justify your response.

10. What is ■ Sectors, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to segments and explain each. Use a simple example.

12. Give two examples related to combinations of plane figures. and explain each. Use a school-life example.

13. Give two examples related to ■ Sectors and explain each. Use a real-world example.

14. Give two examples related to segments and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to combinations of plane figures. and explain each. Explain in your own words.

16. Give two examples related to ■ Sectors and explain each. Give a step-by-step response.

17. Give two examples related to segments and explain each. Mention one common misconception.

18. Give two examples related to combinations of plane figures. and explain each. State one practical application.

19. Give two examples related to ■ Sectors and explain each. Justify your response.

20. Give two examples related to segments and explain each. Include the relevant rule or principle.

Comparison

21. How is combinations of plane figures. related to ■ Sectors? Compare the two ideas. Use a simple example.

22. How is ■ Sectors related to segments? Compare the two ideas. Use a school-life example.

23. How is segments related to combinations of plane figures.? Compare the two ideas. Use a real-world example.

24. How is combinations of plane figures. related to ■ Sectors? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Sectors related to segments? Compare the two ideas. Explain in your own words.

26. How is segments related to combinations of plane figures.? Compare the two ideas. Give a step-by-step response.

27. How is combinations of plane figures. related to ■ Sectors? Compare the two ideas. Mention one common misconception.

28. How is ■ Sectors related to segments? Compare the two ideas. State one practical application.

29. How is segments related to combinations of plane figures.? Compare the two ideas. Justify your response.

30. How is combinations of plane figures. related to ■ Sectors? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Sectors to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply segments to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply combinations of plane figures. to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply ■ Sectors to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply segments to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply combinations of plane figures. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Sectors to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply segments to a realistic situation. What would you do or calculate? State one practical application.

39. Apply combinations of plane figures. to a realistic situation. What would you do or calculate? Justify your response.

40. Apply ■ Sectors to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of segments. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of combinations of plane figures.. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of ■ Sectors. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of segments. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of combinations of plane figures.. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of ■ Sectors. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of segments. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of combinations of plane figures.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Sectors. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of segments. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show combinations of plane figures.. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Sectors. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show segments. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show combinations of plane figures.. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Sectors. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show segments. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show combinations of plane figures.. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Sectors. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show segments. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show combinations of plane figures.. Include the relevant rule or principle.

MATHEMATICS — 12. Surface Areas and Volumes

Coverage: ■ Combination of solids; conversion; frustum; applications.

Understanding

1. What is ■ Combination of solids, and what are its main features? Use a simple example.

2. What is conversion, and what are its main features? Use a school-life example.

3. What is frustum, and what are its main features? Use a real-world example.

4. What is applications., and what are its main features? Show the idea with a diagram where appropriate.

5. What is ■ Combination of solids, and what are its main features? Explain in your own words.

6. What is conversion, and what are its main features? Give a step-by-step response.

7. What is frustum, and what are its main features? Mention one common misconception.

8. What is applications., and what are its main features? State one practical application.

9. What is ■ Combination of solids, and what are its main features? Justify your response.

10. What is conversion, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to frustum and explain each. Use a simple example.

12. Give two examples related to applications. and explain each. Use a school-life example.

13. Give two examples related to ■ Combination of solids and explain each. Use a real-world example.

14. Give two examples related to conversion and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to frustum and explain each. Explain in your own words.

16. Give two examples related to applications. and explain each. Give a step-by-step response.

17. Give two examples related to ■ Combination of solids and explain each. Mention one common misconception.

18. Give two examples related to conversion and explain each. State one practical application.

19. Give two examples related to frustum and explain each. Justify your response.

20. Give two examples related to applications. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Combination of solids related to conversion? Compare the two ideas. Use a simple example.

22. How is conversion related to frustum? Compare the two ideas. Use a school-life example.

23. How is frustum related to applications.? Compare the two ideas. Use a real-world example.

24. How is applications. related to ■ Combination of solids? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Combination of solids related to conversion? Compare the two ideas. Explain in your own words.

26. How is conversion related to frustum? Compare the two ideas. Give a step-by-step response.

27. How is frustum related to applications.? Compare the two ideas. Mention one common misconception.

28. How is applications. related to ■ Combination of solids? Compare the two ideas. State one practical application.

29. How is ■ Combination of solids related to conversion? Compare the two ideas. Justify your response.

30. How is conversion related to frustum? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply frustum to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply applications. to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply ■ Combination of solids to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply conversion to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply frustum to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply applications. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Combination of solids to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply conversion to a realistic situation. What would you do or calculate? State one practical application.

39. Apply frustum to a realistic situation. What would you do or calculate? Justify your response.

40. Apply applications. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Combination of solids. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of conversion. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of frustum. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of applications.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of ■ Combination of solids. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of conversion. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of frustum. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of applications.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Combination of solids. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of conversion. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show frustum. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Combination of solids. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show conversion. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show frustum. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Combination of solids. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show conversion. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show frustum. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Include the relevant rule or principle.

MATHEMATICS — 13. Statistics

Coverage: ■ Grouped data; mean; median; mode; cumulative frequency/ogives.

Understanding

1. What is ■ Grouped data, and what are its main features? Use a simple example.

2. What is mean, and what are its main features? Use a school-life example.

3. What is median, and what are its main features? Use a real-world example.

4. What is mode, and what are its main features? Show the idea with a diagram where appropriate.

5. What is cumulative frequency/ogives., and what are its main features? Explain in your own words.

6. What is ■ Grouped data, and what are its main features? Give a step-by-step response.

7. What is mean, and what are its main features? Mention one common misconception.

8. What is median, and what are its main features? State one practical application.

9. What is mode, and what are its main features? Justify your response.

10. What is cumulative frequency/ogives., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Grouped data and explain each. Use a simple example.

12. Give two examples related to mean and explain each. Use a school-life example.

13. Give two examples related to median and explain each. Use a real-world example.

14. Give two examples related to mode and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to cumulative frequency/ogives. and explain each. Explain in your own words.

16. Give two examples related to ■ Grouped data and explain each. Give a step-by-step response.

17. Give two examples related to mean and explain each. Mention one common misconception.

18. Give two examples related to median and explain each. State one practical application.

19. Give two examples related to mode and explain each. Justify your response.

20. Give two examples related to cumulative frequency/ogives. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Grouped data related to mean? Compare the two ideas. Use a simple example.

22. How is mean related to median? Compare the two ideas. Use a school-life example.

23. How is median related to mode? Compare the two ideas. Use a real-world example.

24. How is mode related to cumulative frequency/ogives.? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is cumulative frequency/ogives. related to ■ Grouped data? Compare the two ideas. Explain in your own words.

26. How is ■ Grouped data related to mean? Compare the two ideas. Give a step-by-step response.

27. How is mean related to median? Compare the two ideas. Mention one common misconception.

28. How is median related to mode? Compare the two ideas. State one practical application.

29. How is mode related to cumulative frequency/ogives.? Compare the two ideas. Justify your response.

30. How is cumulative frequency/ogives. related to ■ Grouped data? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Grouped data to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply mean to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply median to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply mode to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply cumulative frequency/ogives. to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply ■ Grouped data to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply mean to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply median to a realistic situation. What would you do or calculate? State one practical application.

39. Apply mode to a realistic situation. What would you do or calculate? Justify your response.

40. Apply cumulative frequency/ogives. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Grouped data. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of mean. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of median. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of mode. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of cumulative frequency/ogives.. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of ■ Grouped data. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of mean. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of median. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of mode. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of cumulative frequency/ogives.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Grouped data. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show mean. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show median. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show mode. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show cumulative frequency/ogives.. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Grouped data. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show mean. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show median. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show mode. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show cumulative frequency/ogives.. Include the relevant rule or principle.

MATHEMATICS — 14. Probability

Coverage: ■ Theoretical probability; sample space; simple events.

Understanding

1. What is ■ Theoretical probability, and what are its main features? Use a simple example.

2. What is sample space, and what are its main features? Use a school-life example.

3. What is simple events., and what are its main features? Use a real-world example.

4. What is ■ Theoretical probability, and what are its main features? Show the idea with a diagram where appropriate.

5. What is sample space, and what are its main features? Explain in your own words.

6. What is simple events., and what are its main features? Give a step-by-step response.

7. What is ■ Theoretical probability, and what are its main features? Mention one common misconception.

8. What is sample space, and what are its main features? State one practical application.

9. What is simple events., and what are its main features? Justify your response.

10. What is ■ Theoretical probability, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to sample space and explain each. Use a simple example.

12. Give two examples related to simple events. and explain each. Use a school-life example.

13. Give two examples related to ■ Theoretical probability and explain each. Use a real-world example.

14. Give two examples related to sample space and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to simple events. and explain each. Explain in your own words.

16. Give two examples related to ■ Theoretical probability and explain each. Give a step-by-step response.

17. Give two examples related to sample space and explain each. Mention one common misconception.

18. Give two examples related to simple events. and explain each. State one practical application.

19. Give two examples related to ■ Theoretical probability and explain each. Justify your response.

20. Give two examples related to sample space and explain each. Include the relevant rule or principle.

Comparison

21. How is simple events. related to ■ Theoretical probability? Compare the two ideas. Use a simple example.

22. How is ■ Theoretical probability related to sample space? Compare the two ideas. Use a school-life example.

23. How is sample space related to simple events.? Compare the two ideas. Use a real-world example.

24. How is simple events. related to ■ Theoretical probability? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Theoretical probability related to sample space? Compare the two ideas. Explain in your own words.

26. How is sample space related to simple events.? Compare the two ideas. Give a step-by-step response.

27. How is simple events. related to ■ Theoretical probability? Compare the two ideas. Mention one common misconception.

28. How is ■ Theoretical probability related to sample space? Compare the two ideas. State one practical application.

29. How is sample space related to simple events.? Compare the two ideas. Justify your response.

30. How is simple events. related to ■ Theoretical probability? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Theoretical probability to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply sample space to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply simple events. to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply ■ Theoretical probability to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply sample space to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply simple events. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Theoretical probability to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply sample space to a realistic situation. What would you do or calculate? State one practical application.

39. Apply simple events. to a realistic situation. What would you do or calculate? Justify your response.

40. Apply ■ Theoretical probability to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of sample space. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of simple events.. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of ■ Theoretical probability. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of sample space. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of simple events.. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of ■ Theoretical probability. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of sample space. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of simple events.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Theoretical probability. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of sample space. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show simple events.. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Theoretical probability. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show sample space. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show simple events.. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Theoretical probability. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show sample space. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show simple events.. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Theoretical probability. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show sample space. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show simple events.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 1. Chemical Reactions and Equations

Coverage: ■ Types; balancing; oxidation/reduction; reaction evidence.

Understanding

1. What is ■ Types, and what are its main features? Use a simple example.

2. What is balancing, and what are its main features? Use a school-life example.

3. What is oxidation/reduction, and what are its main features? Use a real-world example.

4. What is reaction evidence., and what are its main features? Show the idea with a diagram where appropriate.

5. What is ■ Types, and what are its main features? Explain in your own words.

6. What is balancing, and what are its main features? Give a step-by-step response.

7. What is oxidation/reduction, and what are its main features? Mention one common misconception.

8. What is reaction evidence., and what are its main features? State one practical application.

9. What is ■ Types, and what are its main features? Justify your response.

10. What is balancing, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to oxidation/reduction and explain each. Use a simple example.

12. Give two examples related to reaction evidence. and explain each. Use a school-life example.

13. Give two examples related to ■ Types and explain each. Use a real-world example.

14. Give two examples related to balancing and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to oxidation/reduction and explain each. Explain in your own words.

16. Give two examples related to reaction evidence. and explain each. Give a step-by-step response.

17. Give two examples related to ■ Types and explain each. Mention one common misconception.

18. Give two examples related to balancing and explain each. State one practical application.

19. Give two examples related to oxidation/reduction and explain each. Justify your response.

20. Give two examples related to reaction evidence. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Types related to balancing? Compare the two ideas. Use a simple example.

22. How is balancing related to oxidation/reduction? Compare the two ideas. Use a school-life example.

23. How is oxidation/reduction related to reaction evidence.? Compare the two ideas. Use a real-world example.

24. How is reaction evidence. related to ■ Types? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Types related to balancing? Compare the two ideas. Explain in your own words.

26. How is balancing related to oxidation/reduction? Compare the two ideas. Give a step-by-step response.

27. How is oxidation/reduction related to reaction evidence.? Compare the two ideas. Mention one common misconception.

28. How is reaction evidence. related to ■ Types? Compare the two ideas. State one practical application.

29. How is ■ Types related to balancing? Compare the two ideas. Justify your response.

30. How is balancing related to oxidation/reduction? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply oxidation/reduction to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply reaction evidence. to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply ■ Types to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply balancing to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply oxidation/reduction to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply reaction evidence. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Types to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply balancing to a realistic situation. What would you do or calculate? State one practical application.

39. Apply oxidation/reduction to a realistic situation. What would you do or calculate? Justify your response.

40. Apply reaction evidence. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Types. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of balancing. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of oxidation/reduction. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of reaction evidence.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of ■ Types. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of balancing. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of oxidation/reduction. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of reaction evidence.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Types. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of balancing. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show oxidation/reduction. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show reaction evidence.. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Types. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show balancing. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show oxidation/reduction. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show reaction evidence.. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Types. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show balancing. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show oxidation/reduction. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show reaction evidence.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 2. Acids, Bases and Salts

Coverage: ■ Properties; indicators; pH; neutralisation; common salts.

Understanding

1. What is ■ Properties, and what are its main features? Use a simple example.

2. What is indicators, and what are its main features? Use a school-life example.

3. What is pH, and what are its main features? Use a real-world example.

4. What is neutralisation, and what are its main features? Show the idea with a diagram where appropriate.

5. What is common salts., and what are its main features? Explain in your own words.

6. What is ■ Properties, and what are its main features? Give a step-by-step response.

7. What is indicators, and what are its main features? Mention one common misconception.

8. What is pH, and what are its main features? State one practical application.

9. What is neutralisation, and what are its main features? Justify your response.

10. What is common salts., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Properties and explain each. Use a simple example.

12. Give two examples related to indicators and explain each. Use a school-life example.

13. Give two examples related to pH and explain each. Use a real-world example.

14. Give two examples related to neutralisation and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to common salts. and explain each. Explain in your own words.

16. Give two examples related to ■ Properties and explain each. Give a step-by-step response.

17. Give two examples related to indicators and explain each. Mention one common misconception.

18. Give two examples related to pH and explain each. State one practical application.

19. Give two examples related to neutralisation and explain each. Justify your response.

20. Give two examples related to common salts. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Properties related to indicators? Compare the two ideas. Use a simple example.

22. How is indicators related to pH? Compare the two ideas. Use a school-life example.

23. How is pH related to neutralisation? Compare the two ideas. Use a real-world example.

24. How is neutralisation related to common salts.? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is common salts. related to ■ Properties? Compare the two ideas. Explain in your own words.

26. How is ■ Properties related to indicators? Compare the two ideas. Give a step-by-step response.

27. How is indicators related to pH? Compare the two ideas. Mention one common misconception.

28. How is pH related to neutralisation? Compare the two ideas. State one practical application.

29. How is neutralisation related to common salts.? Compare the two ideas. Justify your response.

30. How is common salts. related to ■ Properties? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Properties to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply indicators to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply pH to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply neutralisation to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply common salts. to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply ■ Properties to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply indicators to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply pH to a realistic situation. What would you do or calculate? State one practical application.

39. Apply neutralisation to a realistic situation. What would you do or calculate? Justify your response.

40. Apply common salts. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Properties. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of indicators. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of pH. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of neutralisation. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of common salts.. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of ■ Properties. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of indicators. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of pH. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of neutralisation. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of common salts.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Properties. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show indicators. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show pH. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show neutralisation. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show common salts.. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Properties. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show indicators. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show pH. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show neutralisation. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show common salts.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 3. Metals and Non-metals

Coverage: ■ Properties; reactivity series; extraction; corrosion and prevention.

Understanding

1. What is ■ Properties, and what are its main features? Use a simple example.

2. What is reactivity series, and what are its main features? Use a school-life example.

3. What is extraction, and what are its main features? Use a real-world example.

4. What is corrosion and prevention., and what are its main features? Show the idea with a diagram where appropriate.

5. What is ■ Properties, and what are its main features? Explain in your own words.

6. What is reactivity series, and what are its main features? Give a step-by-step response.

7. What is extraction, and what are its main features? Mention one common misconception.

8. What is corrosion and prevention., and what are its main features? State one practical application.

9. What is ■ Properties, and what are its main features? Justify your response.

10. What is reactivity series, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to extraction and explain each. Use a simple example.

12. Give two examples related to corrosion and prevention. and explain each. Use a school-life example.

13. Give two examples related to ■ Properties and explain each. Use a real-world example.

14. Give two examples related to reactivity series and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to extraction and explain each. Explain in your own words.

16. Give two examples related to corrosion and prevention. and explain each. Give a step-by-step response.

17. Give two examples related to ■ Properties and explain each. Mention one common misconception.

18. Give two examples related to reactivity series and explain each. State one practical application.

19. Give two examples related to extraction and explain each. Justify your response.

20. Give two examples related to corrosion and prevention. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Properties related to reactivity series? Compare the two ideas. Use a simple example.

22. How is reactivity series related to extraction? Compare the two ideas. Use a school-life example.

23. How is extraction related to corrosion and prevention.? Compare the two ideas. Use a real-world example.

24. How is corrosion and prevention. related to ■ Properties? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Properties related to reactivity series? Compare the two ideas. Explain in your own words.

26. How is reactivity series related to extraction? Compare the two ideas. Give a step-by-step response.

27. How is extraction related to corrosion and prevention.? Compare the two ideas. Mention one common misconception.

28. How is corrosion and prevention. related to ■ Properties? Compare the two ideas. State one practical application.

29. How is ■ Properties related to reactivity series? Compare the two ideas. Justify your response.

30. How is reactivity series related to extraction? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply extraction to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply corrosion and prevention. to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply ■ Properties to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply reactivity series to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply extraction to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply corrosion and prevention. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Properties to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply reactivity series to a realistic situation. What would you do or calculate? State one practical application.

39. Apply extraction to a realistic situation. What would you do or calculate? Justify your response.

40. Apply corrosion and prevention. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Properties. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of reactivity series. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of extraction. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of corrosion and prevention.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of ■ Properties. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of reactivity series. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of extraction. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of corrosion and prevention.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Properties. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of reactivity series. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show extraction. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show corrosion and prevention.. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Properties. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show reactivity series. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show extraction. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show corrosion and prevention.. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Properties. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show reactivity series. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show extraction. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show corrosion and prevention.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 4. Carbon and its Compounds

Coverage: NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus

Understanding

1. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Use a simple example.

2. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Use a school-life example.

3. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Use a real-world example.

4. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Show the idea with a diagram where appropriate.

5. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Explain in your own words.

6. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Give a step-by-step response.

7. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Mention one common misconception.

8. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? State one practical application.

9. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Justify your response.

10. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Use a simple example.

12. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Use a school-life example.

13. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Use a real-world example.

14. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Explain in your own words.

16. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Give a step-by-step response.

17. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Mention one common misconception.

18. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. State one practical application.

19. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Justify your response.

20. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Include the relevant rule or principle.

Comparison

21. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Use a simple example.

22. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Use a school-life example.

23. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Use a real-world example.

24. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Explain in your own words.

26. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Give a step-by-step response.

27. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Mention one common misconception.

28. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. State one practical application.

29. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Justify your response.

30. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? State one practical application.

39. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Justify your response.

40. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 5. Life Processes

Coverage: ■ Nutrition; respiration; transportation; excretion.

Understanding

1. What is ■ Nutrition, and what are its main features? Use a simple example.

2. What is respiration, and what are its main features? Use a school-life example.

3. What is transportation, and what are its main features? Use a real-world example.

4. What is excretion., and what are its main features? Show the idea with a diagram where appropriate.

5. What is ■ Nutrition, and what are its main features? Explain in your own words.

6. What is respiration, and what are its main features? Give a step-by-step response.

7. What is transportation, and what are its main features? Mention one common misconception.

8. What is excretion., and what are its main features? State one practical application.

9. What is ■ Nutrition, and what are its main features? Justify your response.

10. What is respiration, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to transportation and explain each. Use a simple example.

12. Give two examples related to excretion. and explain each. Use a school-life example.

13. Give two examples related to ■ Nutrition and explain each. Use a real-world example.

14. Give two examples related to respiration and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to transportation and explain each. Explain in your own words.

16. Give two examples related to excretion. and explain each. Give a step-by-step response.

17. Give two examples related to ■ Nutrition and explain each. Mention one common misconception.

18. Give two examples related to respiration and explain each. State one practical application.

19. Give two examples related to transportation and explain each. Justify your response.

20. Give two examples related to excretion. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Nutrition related to respiration? Compare the two ideas. Use a simple example.

22. How is respiration related to transportation? Compare the two ideas. Use a school-life example.

23. How is transportation related to excretion.? Compare the two ideas. Use a real-world example.

24. How is excretion. related to ■ Nutrition? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Nutrition related to respiration? Compare the two ideas. Explain in your own words.

26. How is respiration related to transportation? Compare the two ideas. Give a step-by-step response.

27. How is transportation related to excretion.? Compare the two ideas. Mention one common misconception.

28. How is excretion. related to ■ Nutrition? Compare the two ideas. State one practical application.

29. How is ■ Nutrition related to respiration? Compare the two ideas. Justify your response.

30. How is respiration related to transportation? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply transportation to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply excretion. to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply ■ Nutrition to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply respiration to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply transportation to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply excretion. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Nutrition to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply respiration to a realistic situation. What would you do or calculate? State one practical application.

39. Apply transportation to a realistic situation. What would you do or calculate? Justify your response.

40. Apply excretion. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Nutrition. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of respiration. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of transportation. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of excretion.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of ■ Nutrition. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of respiration. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of transportation. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of excretion.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Nutrition. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of respiration. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show transportation. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show excretion.. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Nutrition. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show respiration. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show transportation. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show excretion.. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Nutrition. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show respiration. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show transportation. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show excretion.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 6. Control and Coordination

Coverage: ■ Nervous system; reflexes; hormones; plant coordination.

Understanding

1. What is ■ Nervous system, and what are its main features? Use a simple example.

2. What is reflexes, and what are its main features? Use a school-life example.

3. What is hormones, and what are its main features? Use a real-world example.

4. What is plant coordination., and what are its main features? Show the idea with a diagram where appropriate.

5. What is ■ Nervous system, and what are its main features? Explain in your own words.

6. What is reflexes, and what are its main features? Give a step-by-step response.

7. What is hormones, and what are its main features? Mention one common misconception.

8. What is plant coordination., and what are its main features? State one practical application.

9. What is ■ Nervous system, and what are its main features? Justify your response.

10. What is reflexes, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to hormones and explain each. Use a simple example.

12. Give two examples related to plant coordination. and explain each. Use a school-life example.

13. Give two examples related to ■ Nervous system and explain each. Use a real-world example.

14. Give two examples related to reflexes and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to hormones and explain each. Explain in your own words.

16. Give two examples related to plant coordination. and explain each. Give a step-by-step response.

17. Give two examples related to ■ Nervous system and explain each. Mention one common misconception.

18. Give two examples related to reflexes and explain each. State one practical application.

19. Give two examples related to hormones and explain each. Justify your response.

20. Give two examples related to plant coordination. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Nervous system related to reflexes? Compare the two ideas. Use a simple example.

22. How is reflexes related to hormones? Compare the two ideas. Use a school-life example.

23. How is hormones related to plant coordination.? Compare the two ideas. Use a real-world example.

24. How is plant coordination. related to ■ Nervous system? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Nervous system related to reflexes? Compare the two ideas. Explain in your own words.

26. How is reflexes related to hormones? Compare the two ideas. Give a step-by-step response.

27. How is hormones related to plant coordination.? Compare the two ideas. Mention one common misconception.

28. How is plant coordination. related to ■ Nervous system? Compare the two ideas. State one practical application.

29. How is ■ Nervous system related to reflexes? Compare the two ideas. Justify your response.

30. How is reflexes related to hormones? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply hormones to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply plant coordination. to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply ■ Nervous system to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply reflexes to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply hormones to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply plant coordination. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Nervous system to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply reflexes to a realistic situation. What would you do or calculate? State one practical application.

39. Apply hormones to a realistic situation. What would you do or calculate? Justify your response.

40. Apply plant coordination. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Nervous system. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of reflexes. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of hormones. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of plant coordination.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of ■ Nervous system. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of reflexes. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of hormones. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of plant coordination.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Nervous system. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of reflexes. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show hormones. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show plant coordination.. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Nervous system. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show reflexes. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show hormones. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show plant coordination.. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Nervous system. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show reflexes. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show hormones. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show plant coordination.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 7. How do Organisms Reproduce?

Coverage: ■ Asexual/sexual reproduction; reproductive systems; reproductive health.

Understanding

1. What is ■ Asexual/sexual reproduction, and what are its main features? Use a simple example.

2. What is reproductive systems, and what are its main features? Use a school-life example.

3. What is reproductive health., and what are its main features? Use a real-world example.

4. What is ■ Asexual/sexual reproduction, and what are its main features? Show the idea with a diagram where appropriate.

5. What is reproductive systems, and what are its main features? Explain in your own words.

6. What is reproductive health., and what are its main features? Give a step-by-step response.

7. What is ■ Asexual/sexual reproduction, and what are its main features? Mention one common misconception.

8. What is reproductive systems, and what are its main features? State one practical application.

9. What is reproductive health., and what are its main features? Justify your response.

10. What is ■ Asexual/sexual reproduction, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to reproductive systems and explain each. Use a simple example.

12. Give two examples related to reproductive health. and explain each. Use a school-life example.

13. Give two examples related to ■ Asexual/sexual reproduction and explain each. Use a real-world example.

14. Give two examples related to reproductive systems and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to reproductive health. and explain each. Explain in your own words.

16. Give two examples related to ■ Asexual/sexual reproduction and explain each. Give a step-by-step response.

17. Give two examples related to reproductive systems and explain each. Mention one common misconception.

18. Give two examples related to reproductive health. and explain each. State one practical application.

19. Give two examples related to ■ Asexual/sexual reproduction and explain each. Justify your response.

20. Give two examples related to reproductive systems and explain each. Include the relevant rule or principle.

Comparison

21. How is reproductive health. related to ■ Asexual/sexual reproduction? Compare the two ideas. Use a simple example.

22. How is ■ Asexual/sexual reproduction related to reproductive systems? Compare the two ideas. Use a school-life example.

23. How is reproductive systems related to reproductive health.? Compare the two ideas. Use a real-world example.

24. How is reproductive health. related to ■ Asexual/sexual reproduction? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Asexual/sexual reproduction related to reproductive systems? Compare the two ideas. Explain in your own words.

26. How is reproductive systems related to reproductive health.? Compare the two ideas. Give a step-by-step response.

27. How is reproductive health. related to ■ Asexual/sexual reproduction? Compare the two ideas. Mention one common misconception.

28. How is ■ Asexual/sexual reproduction related to reproductive systems? Compare the two ideas. State one practical application.

29. How is reproductive systems related to reproductive health.? Compare the two ideas. Justify your response.

30. How is reproductive health. related to ■ Asexual/sexual reproduction? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Asexual/sexual reproduction to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply reproductive systems to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply reproductive health. to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply ■ Asexual/sexual reproduction to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply reproductive systems to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply reproductive health. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Asexual/sexual reproduction to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply reproductive systems to a realistic situation. What would you do or calculate? State one practical application.

39. Apply reproductive health. to a realistic situation. What would you do or calculate? Justify your response.

40. Apply ■ Asexual/sexual reproduction to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of reproductive systems. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of reproductive health.. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of ■ Asexual/sexual reproduction. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of reproductive systems. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of reproductive health.. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of ■ Asexual/sexual reproduction. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of reproductive systems. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of reproductive health.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Asexual/sexual reproduction. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of reproductive systems. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show reproductive health.. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Asexual/sexual reproduction. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show reproductive systems. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show reproductive health.. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Asexual/sexual reproduction. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show reproductive systems. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show reproductive health.. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Asexual/sexual reproduction. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show reproductive systems. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show reproductive health.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 8. Heredity

Coverage: ■ Mendelian inheritance; variation; sex determination; evolution.

Understanding

1. What is ■ Mendelian inheritance, and what are its main features? Use a simple example.

2. What is variation, and what are its main features? Use a school-life example.

3. What is sex determination, and what are its main features? Use a real-world example.

4. What is evolution., and what are its main features? Show the idea with a diagram where appropriate.

5. What is ■ Mendelian inheritance, and what are its main features? Explain in your own words.

6. What is variation, and what are its main features? Give a step-by-step response.

7. What is sex determination, and what are its main features? Mention one common misconception.

8. What is evolution., and what are its main features? State one practical application.

9. What is ■ Mendelian inheritance, and what are its main features? Justify your response.

10. What is variation, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to sex determination and explain each. Use a simple example.

12. Give two examples related to evolution. and explain each. Use a school-life example.

13. Give two examples related to ■ Mendelian inheritance and explain each. Use a real-world example.

14. Give two examples related to variation and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to sex determination and explain each. Explain in your own words.

16. Give two examples related to evolution. and explain each. Give a step-by-step response.

17. Give two examples related to ■ Mendelian inheritance and explain each. Mention one common misconception.

18. Give two examples related to variation and explain each. State one practical application.

19. Give two examples related to sex determination and explain each. Justify your response.

20. Give two examples related to evolution. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Mendelian inheritance related to variation? Compare the two ideas. Use a simple example.

22. How is variation related to sex determination? Compare the two ideas. Use a school-life example.

23. How is sex determination related to evolution.? Compare the two ideas. Use a real-world example.

24. How is evolution. related to ■ Mendelian inheritance? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Mendelian inheritance related to variation? Compare the two ideas. Explain in your own words.

26. How is variation related to sex determination? Compare the two ideas. Give a step-by-step response.

27. How is sex determination related to evolution.? Compare the two ideas. Mention one common misconception.

28. How is evolution. related to ■ Mendelian inheritance? Compare the two ideas. State one practical application.

29. How is ■ Mendelian inheritance related to variation? Compare the two ideas. Justify your response.

30. How is variation related to sex determination? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply sex determination to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply evolution. to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply ■ Mendelian inheritance to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply variation to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply sex determination to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply evolution. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Mendelian inheritance to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply variation to a realistic situation. What would you do or calculate? State one practical application.

39. Apply sex determination to a realistic situation. What would you do or calculate? Justify your response.

40. Apply evolution. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Mendelian inheritance. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of variation. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of sex determination. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of evolution.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of ■ Mendelian inheritance. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of variation. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of sex determination. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of evolution.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Mendelian inheritance. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of variation. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show sex determination. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show evolution.. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Mendelian inheritance. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show variation. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show sex determination. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show evolution.. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Mendelian inheritance. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show variation. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show sex determination. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show evolution.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 9. Light — Reflection and Refraction

Coverage: ■ Mirrors; lenses; ray diagrams; refraction; lens formula applications.

Understanding

1. What is ■ Mirrors, and what are its main features? Use a simple example.

2. What is lenses, and what are its main features? Use a school-life example.

3. What is ray diagrams, and what are its main features? Use a real-world example.

4. What is refraction, and what are its main features? Show the idea with a diagram where appropriate.

5. What is lens formula applications., and what are its main features? Explain in your own words.

6. What is ■ Mirrors, and what are its main features? Give a step-by-step response.

7. What is lenses, and what are its main features? Mention one common misconception.

8. What is ray diagrams, and what are its main features? State one practical application.

9. What is refraction, and what are its main features? Justify your response.

10. What is lens formula applications., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Mirrors and explain each. Use a simple example.

12. Give two examples related to lenses and explain each. Use a school-life example.

13. Give two examples related to ray diagrams and explain each. Use a real-world example.

14. Give two examples related to refraction and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to lens formula applications. and explain each. Explain in your own words.

16. Give two examples related to ■ Mirrors and explain each. Give a step-by-step response.

17. Give two examples related to lenses and explain each. Mention one common misconception.

18. Give two examples related to ray diagrams and explain each. State one practical application.

19. Give two examples related to refraction and explain each. Justify your response.

20. Give two examples related to lens formula applications. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Mirrors related to lenses? Compare the two ideas. Use a simple example.

22. How is lenses related to ray diagrams? Compare the two ideas. Use a school-life example.

23. How is ray diagrams related to refraction? Compare the two ideas. Use a real-world example.

24. How is refraction related to lens formula applications.? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is lens formula applications. related to ■ Mirrors? Compare the two ideas. Explain in your own words.

26. How is ■ Mirrors related to lenses? Compare the two ideas. Give a step-by-step response.

27. How is lenses related to ray diagrams? Compare the two ideas. Mention one common misconception.

28. How is ray diagrams related to refraction? Compare the two ideas. State one practical application.

29. How is refraction related to lens formula applications.? Compare the two ideas. Justify your response.

30. How is lens formula applications. related to ■ Mirrors? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Mirrors to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply lenses to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply ray diagrams to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply refraction to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply lens formula applications. to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply ■ Mirrors to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply lenses to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply ray diagrams to a realistic situation. What would you do or calculate? State one practical application.

39. Apply refraction to a realistic situation. What would you do or calculate? Justify your response.

40. Apply lens formula applications. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Mirrors. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of lenses. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of ray diagrams. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of refraction. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of lens formula applications.. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of ■ Mirrors. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of lenses. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of ray diagrams. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of refraction. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of lens formula applications.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Mirrors. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show lenses. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show ray diagrams. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show refraction. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show lens formula applications.. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Mirrors. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show lenses. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show ray diagrams. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show refraction. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show lens formula applications.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 10. The Human Eye and the Colourful World

Coverage: ■ Eye structure; defects; correction; atmospheric refraction; scattering.

Understanding

1. What is ■ Eye structure, and what are its main features? Use a simple example.

2. What is defects, and what are its main features? Use a school-life example.

3. What is correction, and what are its main features? Use a real-world example.

4. What is atmospheric refraction, and what are its main features? Show the idea with a diagram where appropriate.

5. What is scattering., and what are its main features? Explain in your own words.

6. What is ■ Eye structure, and what are its main features? Give a step-by-step response.

7. What is defects, and what are its main features? Mention one common misconception.

8. What is correction, and what are its main features? State one practical application.

9. What is atmospheric refraction, and what are its main features? Justify your response.

10. What is scattering., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Eye structure and explain each. Use a simple example.

12. Give two examples related to defects and explain each. Use a school-life example.

13. Give two examples related to correction and explain each. Use a real-world example.

14. Give two examples related to atmospheric refraction and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to scattering. and explain each. Explain in your own words.

16. Give two examples related to ■ Eye structure and explain each. Give a step-by-step response.

17. Give two examples related to defects and explain each. Mention one common misconception.

18. Give two examples related to correction and explain each. State one practical application.

19. Give two examples related to atmospheric refraction and explain each. Justify your response.

20. Give two examples related to scattering. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Eye structure related to defects? Compare the two ideas. Use a simple example.

22. How is defects related to correction? Compare the two ideas. Use a school-life example.

23. How is correction related to atmospheric refraction? Compare the two ideas. Use a real-world example.

24. How is atmospheric refraction related to scattering.? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is scattering. related to ■ Eye structure? Compare the two ideas. Explain in your own words.

26. How is ■ Eye structure related to defects? Compare the two ideas. Give a step-by-step response.

27. How is defects related to correction? Compare the two ideas. Mention one common misconception.

28. How is correction related to atmospheric refraction? Compare the two ideas. State one practical application.

29. How is atmospheric refraction related to scattering.? Compare the two ideas. Justify your response.

30. How is scattering. related to ■ Eye structure? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Eye structure to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply defects to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply correction to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply atmospheric refraction to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply scattering. to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply ■ Eye structure to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply defects to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply correction to a realistic situation. What would you do or calculate? State one practical application.

39. Apply atmospheric refraction to a realistic situation. What would you do or calculate? Justify your response.

40. Apply scattering. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Eye structure. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of defects. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of correction. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of atmospheric refraction. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of scattering.. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of ■ Eye structure. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of defects. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of correction. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of atmospheric refraction. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of scattering.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Eye structure. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show defects. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show correction. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show atmospheric refraction. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show scattering.. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Eye structure. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show defects. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show correction. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show atmospheric refraction. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show scattering.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 11. Electricity

Coverage: ■ Current; potential difference; resistance; Ohm’s law; circuits; power/energy.

Understanding

1. What is ■ Current, and what are its main features? Use a simple example.

2. What is potential difference, and what are its main features? Use a school-life example.

3. What is resistance, and what are its main features? Use a real-world example.

4. What is Ohm’s law, and what are its main features? Show the idea with a diagram where appropriate.

5. What is circuits, and what are its main features? Explain in your own words.

6. What is power/energy., and what are its main features? Give a step-by-step response.

7. What is ■ Current, and what are its main features? Mention one common misconception.

8. What is potential difference, and what are its main features? State one practical application.

9. What is resistance, and what are its main features? Justify your response.

10. What is Ohm’s law, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to circuits and explain each. Use a simple example.

12. Give two examples related to power/energy. and explain each. Use a school-life example.

13. Give two examples related to ■ Current and explain each. Use a real-world example.

14. Give two examples related to potential difference and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to resistance and explain each. Explain in your own words.

16. Give two examples related to Ohm’s law and explain each. Give a step-by-step response.

17. Give two examples related to circuits and explain each. Mention one common misconception.

18. Give two examples related to power/energy. and explain each. State one practical application.

19. Give two examples related to ■ Current and explain each. Justify your response.

20. Give two examples related to potential difference and explain each. Include the relevant rule or principle.

Comparison

21. How is resistance related to Ohm’s law? Compare the two ideas. Use a simple example.

22. How is Ohm’s law related to circuits? Compare the two ideas. Use a school-life example.

23. How is circuits related to power/energy.? Compare the two ideas. Use a real-world example.

24. How is power/energy. related to ■ Current? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Current related to potential difference? Compare the two ideas. Explain in your own words.

26. How is potential difference related to resistance? Compare the two ideas. Give a step-by-step response.

27. How is resistance related to Ohm’s law? Compare the two ideas. Mention one common misconception.

28. How is Ohm’s law related to circuits? Compare the two ideas. State one practical application.

29. How is circuits related to power/energy.? Compare the two ideas. Justify your response.

30. How is power/energy. related to ■ Current? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Current to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply potential difference to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply resistance to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply Ohm’s law to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply circuits to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply power/energy. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Current to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply potential difference to a realistic situation. What would you do or calculate? State one practical application.

39. Apply resistance to a realistic situation. What would you do or calculate? Justify your response.

40. Apply Ohm’s law to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of circuits. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of power/energy.. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of ■ Current. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of potential difference. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of resistance. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of Ohm’s law. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of circuits. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of power/energy.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Current. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of potential difference. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show resistance. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show Ohm’s law. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show circuits. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show power/energy.. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Current. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show potential difference. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show resistance. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show Ohm’s law. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show circuits. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show power/energy.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 12. Magnetic Effects of Electric Current

Coverage: ■ Magnetic field; force; electromagnetic induction; motor/generator principles.

Understanding

1. What is ■ Magnetic field, and what are its main features? Use a simple example.

2. What is force, and what are its main features? Use a school-life example.

3. What is electromagnetic induction, and what are its main features? Use a real-world example.

4. What is motor/generator principles., and what are its main features? Show the idea with a diagram where appropriate.

5. What is ■ Magnetic field, and what are its main features? Explain in your own words.

6. What is force, and what are its main features? Give a step-by-step response.

7. What is electromagnetic induction, and what are its main features? Mention one common misconception.

8. What is motor/generator principles., and what are its main features? State one practical application.

9. What is ■ Magnetic field, and what are its main features? Justify your response.

10. What is force, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to electromagnetic induction and explain each. Use a simple example.

12. Give two examples related to motor/generator principles. and explain each. Use a school-life example.

13. Give two examples related to ■ Magnetic field and explain each. Use a real-world example.

14. Give two examples related to force and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to electromagnetic induction and explain each. Explain in your own words.

16. Give two examples related to motor/generator principles. and explain each. Give a step-by-step response.

17. Give two examples related to ■ Magnetic field and explain each. Mention one common misconception.

18. Give two examples related to force and explain each. State one practical application.

19. Give two examples related to electromagnetic induction and explain each. Justify your response.

20. Give two examples related to motor/generator principles. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Magnetic field related to force? Compare the two ideas. Use a simple example.

22. How is force related to electromagnetic induction? Compare the two ideas. Use a school-life example.

23. How is electromagnetic induction related to motor/generator principles.? Compare the two ideas. Use a real-world example.

24. How is motor/generator principles. related to ■ Magnetic field? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Magnetic field related to force? Compare the two ideas. Explain in your own words.

26. How is force related to electromagnetic induction? Compare the two ideas. Give a step-by-step response.

27. How is electromagnetic induction related to motor/generator principles.? Compare the two ideas. Mention one common misconception.

28. How is motor/generator principles. related to ■ Magnetic field? Compare the two ideas. State one practical application.

29. How is ■ Magnetic field related to force? Compare the two ideas. Justify your response.

30. How is force related to electromagnetic induction? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply electromagnetic induction to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply motor/generator principles. to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply ■ Magnetic field to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply force to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply electromagnetic induction to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply motor/generator principles. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Magnetic field to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply force to a realistic situation. What would you do or calculate? State one practical application.

39. Apply electromagnetic induction to a realistic situation. What would you do or calculate? Justify your response.

40. Apply motor/generator principles. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Magnetic field. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of force. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of electromagnetic induction. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of motor/generator principles.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of ■ Magnetic field. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of force. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of electromagnetic induction. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of motor/generator principles.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Magnetic field. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of force. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show electromagnetic induction. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show motor/generator principles.. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Magnetic field. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show force. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show electromagnetic induction. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show motor/generator principles.. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Magnetic field. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show force. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show electromagnetic induction. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show motor/generator principles.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 13. Our Environment

Coverage: ■ Ecosystems; food chains/webs; energy flow; environmental management.

Understanding

1. What is ■ Ecosystems, and what are its main features? Use a simple example.

2. What is food chains/webs, and what are its main features? Use a school-life example.

3. What is energy flow, and what are its main features? Use a real-world example.

4. What is environmental management., and what are its main features? Show the idea with a diagram where appropriate.

5. What is ■ Ecosystems, and what are its main features? Explain in your own words.

6. What is food chains/webs, and what are its main features? Give a step-by-step response.

7. What is energy flow, and what are its main features? Mention one common misconception.

8. What is environmental management., and what are its main features? State one practical application.

9. What is ■ Ecosystems, and what are its main features? Justify your response.

10. What is food chains/webs, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to energy flow and explain each. Use a simple example.

12. Give two examples related to environmental management. and explain each. Use a school-life example.

13. Give two examples related to ■ Ecosystems and explain each. Use a real-world example.

14. Give two examples related to food chains/webs and explain each. Show the idea with a diagram where appropriate.

15. Give two examples related to energy flow and explain each. Explain in your own words.

16. Give two examples related to environmental management. and explain each. Give a step-by-step response.

17. Give two examples related to ■ Ecosystems and explain each. Mention one common misconception.

18. Give two examples related to food chains/webs and explain each. State one practical application.

19. Give two examples related to energy flow and explain each. Justify your response.

20. Give two examples related to environmental management. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Ecosystems related to food chains/webs? Compare the two ideas. Use a simple example.

22. How is food chains/webs related to energy flow? Compare the two ideas. Use a school-life example.

23. How is energy flow related to environmental management.? Compare the two ideas. Use a real-world example.

24. How is environmental management. related to ■ Ecosystems? Compare the two ideas. Show the idea with a diagram where appropriate.

25. How is ■ Ecosystems related to food chains/webs? Compare the two ideas. Explain in your own words.

26. How is food chains/webs related to energy flow? Compare the two ideas. Give a step-by-step response.

27. How is energy flow related to environmental management.? Compare the two ideas. Mention one common misconception.

28. How is environmental management. related to ■ Ecosystems? Compare the two ideas. State one practical application.

29. How is ■ Ecosystems related to food chains/webs? Compare the two ideas. Justify your response.

30. How is food chains/webs related to energy flow? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply energy flow to a realistic situation. What would you do or calculate? Use a simple example.

32. Apply environmental management. to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply ■ Ecosystems to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply food chains/webs to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply energy flow to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply environmental management. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Ecosystems to a realistic situation. What would you do or calculate? Mention one common misconception.

38. Apply food chains/webs to a realistic situation. What would you do or calculate? State one practical application.

39. Apply energy flow to a realistic situation. What would you do or calculate? Justify your response.

40. Apply environmental management. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Ecosystems. Identify a likely error and explain the correction. Use a simple example.

42. A student gives an incorrect explanation of food chains/webs. Identify a likely error and explain the correction. Use a school-life example.

43. A student gives an incorrect explanation of energy flow. Identify a likely error and explain the correction. Use a real-world example.

44. A student gives an incorrect explanation of environmental management.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.

45. A student gives an incorrect explanation of ■ Ecosystems. Identify a likely error and explain the correction. Explain in your own words.

46. A student gives an incorrect explanation of food chains/webs. Identify a likely error and explain the correction. Give a step-by-step response.

47. A student gives an incorrect explanation of energy flow. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of environmental management.. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of ■ Ecosystems. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of food chains/webs. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show energy flow. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show environmental management.. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Ecosystems. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show food chains/webs. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show energy flow. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show environmental management.. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Ecosystems. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show food chains/webs. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show energy flow. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show environmental management.. Include the relevant rule or principle.